

Full Report: Refined 3D Index of 5_1^2

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1 Manifold Summary

- Manifold: 5_1^2
- Number of tetrahedra: $n = 4$
- Number of cusps: $r = 2$
- Independent easy edges: 1
- Hard edges (fugacity variables): 1
- q -order (half): 10 (q -order: 10/2)

2 SnaPy Gluing Equations

The gluing equation matrix has rows:

- Rows 0...3: edge equations
- Rows 4,6,...: meridian equations (per cusp)
- Rows 5,7,...: longitude equations (per cusp)

Columns are ordered as $Z_1, Z'_1, Z''_1, Z_2, Z'_2, Z''_2, \dots, Z_4, Z'_4, Z''_4$.

$$G_{\text{glue}} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 2 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 2 & 1 & 0 & 2 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & -1 & -2 & 0 & 0 & -1 & 0 & -2 & -1 & 3 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 & -1 & 0 & 1 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}$$

Cusp Equations

Cusp 0:

$$\begin{aligned} \mu_0 &= -Z''_1 - Z'_3 + Z_4 \\ \lambda_0 &= Z_1 - Z'_1 - 2Z''_1 - Z''_2 - 2Z'_3 - Z''_3 + 3Z_4 - Z'_4 \end{aligned}$$

Cusp 1:

$$\begin{aligned} \mu_1 &= Z_1 - Z'_2 - Z''_3 \\ \lambda_1 &= 2Z_1 - Z'_1 - Z'_2 + Z_3 - Z''_3 - Z_4 - Z''_4 \end{aligned}$$

3 Edge Classification

Total SnaPy edge equations: 4. Independent edge basis rank: $n - r = 2$. Of these, 1 are **easy** and 1 are **hard**.

3.1 Hard Edges

Hard edges contribute fugacity variables $\eta_0, \eta_1, \dots$ to the refined index.

1. $C_{\text{hard},0} = 2Z'_1 + Z''_1 + 2Z'_2 + Z''_2 + Z''_3 + Z''_4$

3.2 Easy Edges (independent basis)

Easy edges are summed over without fugacity (internal charges only).

1. $C_{\text{easy},0} = Z_1 + Z_2 + Z_3 + Z_4$

Basis ordering in g_{NZ} (position block rows): meridians (r rows), hard edges, easy edges.

4 Neumann–Zagier Data

4.1 Symplectic Matrix g_{NZ}

Row ordering: meridians ($\mu_0, \dots, \mu_{r-1}$), hard edges, easy edges | longitudes/2 ($\lambda_0/2, \dots$), Γ vectors.
Column ordering: $Z_1, \dots, Z_n, Z_1'', \dots, Z_n''$ (block form).

$$g_{NZ} = \begin{pmatrix} 0 & 0 & 1 & 1 & -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & -1 & 0 \\ -2 & -2 & 0 & 0 & -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ \frac{3}{2} & \frac{5}{2} & 3 & 2 & -1 & \frac{1}{2} & \frac{3}{2} & 0 \end{pmatrix}$$

4.2 Inverse g_{NZ}^{-1}

$$g_{NZ}^{-1} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -1 & 1 & 0 & 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & -1 & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 & \frac{3}{2} & -1 & 1 & -1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 & -1 & 0 \\ -1 & -\frac{3}{2} & \frac{1}{2} & -\frac{3}{2} & 0 & 1 & -2 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{5}{2} & 0 & 1 & -2 & 1 \\ -1 & -\frac{1}{2} & -\frac{1}{2} & -3 & 1 & 0 & 0 & 1 \\ -2 & \frac{1}{2} & \frac{1}{2} & -2 & 1 & 0 & 0 & 1 \end{pmatrix}$$

4.3 Affine Shifts

$$\nu_x = (-1, -1, 2, -2)$$

$$\nu_p = (-2, -1, 0, 0)$$

5 Local Charges Formula

The refined index is computed by the formula:

$$I^{\text{ref}}(q; \eta_0, \dots, \eta_{k-1}) = \sum_{e_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} \left[\prod_{a=0}^{k-1} \eta_a^{e_{r+a}} \right] \cdot (-q^{1/2})^\varphi \cdot \prod_{j=1}^n I_\Delta((g_{NZ}^{-1}\kappa)_j, (g_{NZ}^{-1}\kappa)_{n+j}) \quad (1)$$

where:

- $\kappa = (m_{\text{ext}}, \underbrace{0, \dots, 0}_{n-r}, e_{\text{ext}}, e_{\text{int}})$ is the $2n$ -vector of external + internal charges.
- The **local charges** (tetrahedron arguments) are $(m_j, e_j) = (g_{NZ}^{-1}\kappa)_j$ and $(g_{NZ}^{-1}\kappa)_{n+j}$.
- The **phase exponent** is $\varphi = m_{\text{full}} \cdot \nu_p - e_{\text{full}} \cdot \nu_x$.
- $I_\Delta(m, e)$ is the tetrahedron index.
- $k = \text{num_hard}$ is the number of hard edges carrying fugacity variables.

6 Non-Closable Cycles & Basis Selection

Cusp 0: non-closable cycles found: $(-1, 0), (1, 0)$. Cusp 1: non-closable cycles found: $(-1, 0), (1, 0)$.

Selected Basis

- Cusp 0: slope $(1, 0) \Rightarrow m = 1, e = 0$ [default]
- Cusp 1: slope $(1, 0) \Rightarrow m = 1, e = 0$ [default]

7 Dehn Filled Unrefined Indices

The unrefined Dehn-filled index $I_{P/Q}(q)$ is computed for each slope tested in the non-closable cycle search. A slope is *non-closable* when $I_{P/Q}(q)$ is stably zero.

Cusp 0

Stably zero (non-closable): $(-1, 0), (1, 0)$.

P	Q	$I_{P/Q}(q)$
-1	1	1
0	1	$46 - q - 2q^2 - 2q^3 - 3q^4 - 2q^5$
1	1	$1 - 2q - 3q^2 + 2q^3 + 8q^4 + 18q^5$

Cusp 1

Stably zero (non-closable): $(-1, 0), (1, 0)$.

P	Q	$I_{P/Q}(q)$
-1	1	1
0	1	$46 - q - 2q^2 - 2q^3 - 3q^4 - 2q^5$
1	1	$1 - 2q - 3q^2 + 2q^3 + 8q^4 + 18q^5$

8 Weyl Symmetry & Dehn Filling Compatibility

Convention: $f(\eta) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, $f(\eta) = f(\eta^{-1})$.

$$a = \left(-\frac{1}{2}\right), \quad b = \left(\frac{1}{2}\right)$$

- $a \in \mathbb{Z}$: $\times$
- $2b \in \mathbb{Z}$: $\checkmark$

Dehn filling not directly compatible: $a \notin \mathbb{Z}$.

Weyl symmetry check: $\times$ (6301/6561 entries pass).

9 Refined Index

Evaluated at q -order (half) = 10, giving series up to q^5 .

Output in **Weyl-manifest form**: $f(m, e) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, where the displayed $I(m, e)$ satisfies $I(\eta) = I(\eta^{-1})$.

$$\begin{aligned}
I(-4, 0, -2, -2) &= q^5 \eta^{-2 v_0} \\
I(-4, 0, -2, -1) &= \eta^{v_0} (q^5 \eta^{-3 v_0}) \\
I(-4, 0, -2, 0) &= \eta^{2 v_0} (q^5 \eta^{-4 v_0}) \\
I(-4, 0, -2, 1) &= \eta^{3 v_0} (q^5 \eta^{-5 v_0}) \\
I(-4, 0, -2, 2) &= \eta^{4 v_0} (q^5 \eta^{-6 v_0}) \\
I(-4, 0, -1, -2) &= \eta^{v_0} (q^4 \eta^{-5 v_0} + q^4 \eta^{-3 v_0} + q^5 \eta^{-5 v_0} + q^5 \eta^{-3 v_0}) \\
I(-4, 0, -1, -1) &= \eta^{2 v_0} (q^4 \eta^{-6 v_0} + q^4 \eta^{-4 v_0} + q^5 \eta^{-6 v_0} + q^5 \eta^{-4 v_0}) \\
I(-4, 0, -1, 0) &= \eta^{3 v_0} (q^4 \eta^{-7 v_0} + q^4 \eta^{-5 v_0} + q^5 \eta^{-7 v_0} + q^5 \eta^{-5 v_0}) \\
I(-4, 0, -1, 1) &= \eta^{4 v_0} (q^4 \eta^{-8 v_0} + q^4 \eta^{-6 v_0} + q^5 \eta^{-8 v_0} + q^5 \eta^{-6 v_0}) \\
I(-4, 0, -1, 2) &= \eta^{5 v_0} (q^4 \eta^{-9 v_0} + q^4 \eta^{-7 v_0} + q^5 \eta^{-9 v_0} + q^5 \eta^{-7 v_0}) \\
I(-4, 0, 0, -2) &= \eta^{2 v_0} (q^2 \eta^{-6 v_0} - q^5 \eta^{-6 v_0}) \\
I(-4, 0, 0, -1) &= \eta^{3 v_0} (q^2 \eta^{-7 v_0}) \\
I(-4, 0, 0, 0) &= \eta^{4 v_0} (q^2 \eta^{-8 v_0}) \\
I(-4, 0, 0, 1) &= \eta^{5 v_0} (q^2 \eta^{-9 v_0}) \\
I(-4, 0, 0, 2) &= \eta^{6 v_0} (q^2 \eta^{-10 v_0} - q^5 \eta^{-10 v_0}) \\
I(-4, 0, 1, -2) &= \eta^{3 v_0} (-q^5 \eta^{-9 v_0} - q^5 \eta^{-7 v_0}) \\
I(-4, 0, 1, -1) &= \eta^{4 v_0} (-q^5 \eta^{-10 v_0} - q^5 \eta^{-8 v_0}) \\
I(-4, 0, 1, 0) &= \eta^{5 v_0} (-q^5 \eta^{-11 v_0} - q^5 \eta^{-9 v_0}) \\
I(-4, 0, 1, 1) &= \eta^{6 v_0} (-q^5 \eta^{-12 v_0} - q^5 \eta^{-10 v_0}) \\
I(-4, 0, 1, 2) &= \eta^{7 v_0} (-q^5 \eta^{-13 v_0} - q^5 \eta^{-11 v_0}) \\
I(-4, -2, 2, 2) &= \eta^{10 v_0} (q^5 \eta^{-16 v_0}) \\
I(-4, +2, 2, -2) &= \eta^{2 v_0} (q^5 \eta^{-8 v_0}) \\
I(-2, -2, -2, -2) &= q^3 - q^4 - 2q^4 \eta^{2 v_0} - 4q^5 - 5q^5 \eta^{2 v_0} \\
I(-2, -2, -2, -1) &= \eta^{v_0} (-q^4 \eta^{v_0} - 2q^5 \eta^{-v_0} - 4q^5 \eta^{v_0}) \\
I(-2, -2, -2, 0) &= \eta^{2 v_0} (-q^4 - 2q^5) \\
I(-2, 0, -2, -2) &= \eta^{-2 v_0} (q^3 - q^4 - 2q^4 \eta^{2 v_0} - 4q^5 - 5q^5 \eta^{2 v_0}) \\
I(-2, 0, -2, -1) &= \eta^{-v_0} (q^2 \eta^{v_0} + q^3 \eta^{-v_0} + 2q^3 \eta^{v_0} - q^4 \eta^{v_0} - q^4 \eta^{3 v_0} - 4q^5 \eta^{-v_0} - 9q^5 \eta^{v_0} - 2q^5 \eta^{3 v_0}) \\
I(-2, 0, -2, 0) &= q^2 + q^3 \eta^{-2 v_0} + 3q^3 + q^3 \eta^{2 v_0} - q^4 \eta^{-2 v_0} + q^4 - q^4 \eta^{2 v_0} - 4q^5 \eta^{-2 v_0} - 8q^5 - 4q^5 \eta^{2 v_0} \\
I(-2, 0, -2, 1) &= \eta^{v_0} (q^2 \eta^{-v_0} + 2q^3 \eta^{-v_0} + q^3 \eta^{v_0} - q^4 \eta^{-3 v_0} - q^4 \eta^{-v_0} - 2q^5 \eta^{-3 v_0} - 9q^5 \eta^{-v_0} - 4q^5 \eta^{v_0}) \\
I(-2, 0, -2, 2) &= \eta^{2 v_0} (q^3 - 2q^4 \eta^{-2 v_0} - q^4 - 5q^5 \eta^{-2 v_0} - 4q^5) \\
I(-2, +2, -2, 0) &= \eta^{-2 v_0} (-q^4 - 2q^5) \\
I(-2, +2, -2, 1) &= \eta^{-v_0} (-q^4 \eta^{-v_0} - 4q^5 \eta^{-v_0} - 2q^5 \eta^{v_0}) \\
I(-2, +2, -2, 2) &= q^3 - 2q^4 \eta^{-2 v_0} - q^4 - 5q^5 \eta^{-2 v_0} - 4q^5 \\
I(-2, -2, -1, -2) &= \eta^{v_0} (-q^4 \eta^{-v_0} - 2q^5 \eta^{-3 v_0} - 4q^5 \eta^{-v_0}) \\
I(-2, -2, -1, -1) &= \eta^{2 v_0} (-q^4 \eta^{-2 v_0} - q^5 \eta^{-4 v_0} - 3q^5 \eta^{-2 v_0}) \\
I(-2, -2, -1, 0) &= \eta^{3 v_0} (-q^4 \eta^{-3 v_0} - 2q^5 \eta^{-3 v_0}) \\
I(-2, -2, -1, 2) &= \eta^{5 v_0} (q^5 \eta^{-3 v_0}) \\
I(-2, 0, -1, -2) &= \eta^{-v_0} (q^2 \eta^{-v_0} - q^3 \eta^{v_0} - 3q^4 \eta^{-v_0} - 2q^4 \eta^{v_0} - q^5 \eta^{-3 v_0} - 8q^5 \eta^{-v_0} - 2q^5 \eta^{v_0}) \\
I(-2, 0, -1, -1) &= q^2 \eta^{-2 v_0} + q^2 + q^3 \eta^{-2 v_0} - q^4 \eta^{-2 v_0} - 3q^4 - q^5 \eta^{-4 v_0} - 8q^5 \eta^{-2 v_0} - 8q^5 \\
I(-2, 0, -1, 0) &= \eta^{v_0} (q^2 \eta^{-3 v_0} + q^2 \eta^{-v_0} + q^3 \eta^{-3 v_0} + q^3 \eta^{-v_0} - q^4 \eta^{-3 v_0} - q^4 \eta^{-v_0} - q^5 \eta^{-5 v_0} - 8q^5 \eta^{-3 v_0} - 8q^5 \eta^{-v_0})
\end{aligned}$$

$$\begin{aligned}
I(-2, 0, -1, 1) &= \eta^{2v_0} (q^2 \eta^{-4v_0} + q^2 \eta^{-2v_0} + q^3 \eta^{-2v_0} - 3q^4 \eta^{-4v_0} - q^4 \eta^{-2v_0} - 8q^5 \eta^{-4v_0} - 8q^5 \eta^{-2v_0} - q^5) \\
I(-2, 0, -1, 2) &= \eta^{3v_0} (q^2 \eta^{-3v_0} - q^3 \eta^{-5v_0} - 2q^4 \eta^{-5v_0} - 3q^4 \eta^{-3v_0} - 2q^5 \eta^{-5v_0} - 8q^5 \eta^{-3v_0} - q^5 \eta^{-v_0}) \\
I(-2, +2, -1, -2) &= \eta^{-3v_0} (q^5 \eta^{-v_0}) \\
I(-2, +2, -1, 0) &= \eta^{-v_0} (-q^4 \eta^{-v_0} - 2q^5 \eta^{-v_0}) \\
I(-2, +2, -1, 1) &= -q^4 \eta^{-2v_0} - 3q^5 \eta^{-2v_0} - q^5 \\
I(-2, +2, -1, 2) &= \eta^{v_0} (-q^4 \eta^{-3v_0} - 4q^5 \eta^{-3v_0} - 2q^5 \eta^{-v_0}) \\
I(-2, -2, 0, -2) &= \eta^{2v_0} (-q^4 \eta^{-4v_0} - 2q^5 \eta^{-4v_0}) \\
I(-2, -2, 0, -1) &= \eta^{3v_0} (-q^4 \eta^{-5v_0} - 2q^5 \eta^{-5v_0}) \\
I(-2, -2, 0, 0) &= \eta^{4v_0} (-q^4 \eta^{-6v_0} - q^5 \eta^{-6v_0}) \\
I(-2, -2, 0, 1) &= \eta^{5v_0} (q^5 \eta^{-7v_0} + q^5 \eta^{-5v_0}) \\
I(-2, -2, 0, 2) &= \eta^{6v_0} (q^4 \eta^{-6v_0} + q^5 \eta^{-6v_0}) \\
I(-2, 0, 0, -2) &= q\eta^{-2v_0} - q^2 \eta^{-2v_0} - 2q^3 \eta^{-2v_0} - q^4 \eta^{-4v_0} - 4q^4 \eta^{-2v_0} - 2q^5 \eta^{-4v_0} - 4q^5 \eta^{-2v_0} + q^5 \\
I(-2, 0, 0, -1) &= \eta^{v_0} (q\eta^{-3v_0} - q^3 \eta^{-3v_0} - q^4 \eta^{-5v_0} - 4q^4 \eta^{-3v_0} - q^4 \eta^{-v_0} - 2q^5 \eta^{-5v_0} - 8q^5 \eta^{-3v_0} - q^5 \eta^{-v_0}) \\
I(-2, 0, 0, 0) &= \eta^{2v_0} (q\eta^{-4v_0} - q^4 \eta^{-6v_0} - 4q^4 \eta^{-4v_0} - q^4 \eta^{-2v_0} - 2q^5 \eta^{-6v_0} - 8q^5 \eta^{-4v_0} - 2q^5 \eta^{-2v_0}) \\
I(-2, 0, 0, 1) &= \eta^{3v_0} (q\eta^{-5v_0} - q^3 \eta^{-5v_0} - q^4 \eta^{-7v_0} - 4q^4 \eta^{-5v_0} - q^4 \eta^{-3v_0} - q^5 \eta^{-7v_0} - 8q^5 \eta^{-5v_0} - 2q^5 \eta^{-3v_0}) \\
I(-2, 0, 0, 2) &= \eta^{4v_0} (q\eta^{-6v_0} - q^2 \eta^{-6v_0} - 2q^3 \eta^{-6v_0} - 4q^4 \eta^{-6v_0} - q^4 \eta^{-4v_0} + q^5 \eta^{-8v_0} - 4q^5 \eta^{-6v_0} - 2q^5 \eta^{-4v_0}) \\
I(-2, +2, 0, -2) &= \eta^{-2v_0} (q^4 \eta^{-2v_0} + q^5 \eta^{-2v_0}) \\
I(-2, +2, 0, -1) &= \eta^{-v_0} (q^5 \eta^{-3v_0} + q^5 \eta^{-v_0}) \\
I(-2, +2, 0, 0) &= -q^4 \eta^{-2v_0} - q^5 \eta^{-2v_0} \\
I(-2, +2, 0, 1) &= \eta^{v_0} (-q^4 \eta^{-3v_0} - 2q^5 \eta^{-3v_0}) \\
I(-2, +2, 0, 2) &= \eta^{2v_0} (-q^4 \eta^{-4v_0} - 2q^5 \eta^{-4v_0}) \\
I(-2, -2, 1, 0) &= \eta^{5v_0} (q^5 \eta^{-9v_0} + q^5 \eta^{-7v_0}) \\
I(-2, -2, 1, 1) &= \eta^{6v_0} (q^4 \eta^{-10v_0} + q^4 \eta^{-8v_0} + q^5 \eta^{-10v_0} + 2q^5 \eta^{-8v_0}) \\
I(-2, -2, 1, 2) &= \eta^{7v_0} (q^3 \eta^{-9v_0} + q^4 \eta^{-9v_0} + q^5 \eta^{-9v_0}) \\
I(-2, 0, 1, -2) &= \eta^{v_0} (-q^3 \eta^{-5v_0} - q^3 \eta^{-3v_0} - q^4 \eta^{-5v_0} + 2q^5 \eta^{-3v_0}) \\
I(-2, 0, 1, -1) &= \eta^{2v_0} (-q^3 \eta^{-6v_0} - q^3 \eta^{-4v_0} - q^4 \eta^{-6v_0} - q^4 \eta^{-4v_0} - q^5 \eta^{-6v_0}) \\
I(-2, 0, 1, 0) &= \eta^{3v_0} (-q^3 \eta^{-7v_0} - q^3 \eta^{-5v_0} - q^4 \eta^{-7v_0} - q^4 \eta^{-5v_0} - q^5 \eta^{-7v_0} - q^5 \eta^{-5v_0}) \\
I(-2, 0, 1, 1) &= \eta^{4v_0} (-q^3 \eta^{-8v_0} - q^3 \eta^{-6v_0} - q^4 \eta^{-8v_0} - q^4 \eta^{-6v_0} - q^5 \eta^{-6v_0}) \\
I(-2, 0, 1, 2) &= \eta^{5v_0} (-q^3 \eta^{-9v_0} - q^3 \eta^{-7v_0} - q^4 \eta^{-7v_0} + 2q^5 \eta^{-9v_0}) \\
I(-2, +2, 1, -2) &= \eta^{-v_0} (q^3 \eta^{-3v_0} + q^4 \eta^{-3v_0} + q^5 \eta^{-3v_0}) \\
I(-2, +2, 1, -1) &= q^4 \eta^{-4v_0} + q^4 \eta^{-2v_0} + 2q^5 \eta^{-4v_0} + q^5 \eta^{-2v_0} \\
I(-2, +2, 1, 0) &= \eta^{v_0} (q^5 \eta^{-5v_0} + q^5 \eta^{-3v_0}) \\
I(-2, -4, 2, 2) &= \eta^{10v_0} (q^5 \eta^{-12v_0}) \\
I(-2, -2, 2, -1) &= \eta^{5v_0} (q^5 \eta^{-9v_0}) \\
I(-2, -2, 2, 0) &= \eta^{6v_0} (q^4 \eta^{-10v_0} + q^5 \eta^{-10v_0}) \\
I(-2, -2, 2, 1) &= \eta^{7v_0} (q^3 \eta^{-11v_0} + q^4 \eta^{-11v_0} + q^5 \eta^{-11v_0}) \\
I(-2, -2, 2, 2) &= \eta^{8v_0} (q^2 \eta^{-12v_0} - 2q^5 \eta^{-12v_0}) \\
I(-2, 0, 2, -2) &= \eta^{2v_0} (q^5 \eta^{-6v_0}) \\
I(-2, 0, 2, -1) &= \eta^{3v_0} (q^5 \eta^{-7v_0}) \\
I(-2, 0, 2, 0) &= \eta^{4v_0} (q^5 \eta^{-8v_0}) \\
I(-2, 0, 2, 1) &= \eta^{5v_0} (q^5 \eta^{-9v_0})
\end{aligned}$$

$$\begin{aligned}
I(-2, 0, 2, 2) &= \eta^{6v_0} (q^5 \eta^{-10v_0}) \\
I(-2, +2, 2, -2) &= q^2 \eta^{-4v_0} - 2q^5 \eta^{-4v_0} \\
I(-2, +2, 2, -1) &= \eta^{v_0} (q^3 \eta^{-5v_0} + q^4 \eta^{-5v_0} + q^5 \eta^{-5v_0}) \\
I(-2, +2, 2, 0) &= \eta^{2v_0} (q^4 \eta^{-6v_0} + q^5 \eta^{-6v_0}) \\
I(-2, +2, 2, 1) &= \eta^{3v_0} (q^5 \eta^{-7v_0}) \\
I(-2, +4, 2, -2) &= \eta^{-2v_0} (q^5 \eta^{-4v_0}) \\
I(0, -4, -2, -2) &= q^5 \eta^{6v_0} \\
I(0, -4, -2, -1) &= \eta^{v_0} (q^4 \eta^{5v_0} + q^4 \eta^{7v_0} + q^5 \eta^{5v_0} + q^5 \eta^{7v_0}) \\
I(0, -4, -2, 0) &= \eta^{2v_0} (q^2 \eta^{6v_0} - q^5 \eta^{6v_0}) \\
I(0, -4, -2, 1) &= \eta^{3v_0} (-q^5 \eta^{5v_0} - q^5 \eta^{7v_0}) \\
I(0, -2, -2, -2) &= \eta^{-2v_0} (q^3 \eta^{4v_0} - q^4 \eta^{4v_0} - 2q^4 \eta^{6v_0} - 4q^5 \eta^{4v_0} - 5q^5 \eta^{6v_0}) \\
I(0, -2, -2, -1) &= \eta^{-v_0} (q^2 \eta^{5v_0} - q^3 \eta^{7v_0} - 3q^4 \eta^{5v_0} - 2q^4 \eta^{7v_0} - q^5 \eta^{3v_0} - 8q^5 \eta^{5v_0} - 2q^5 \eta^{7v_0}) \\
I(0, -2, -2, 0) &= q\eta^{6v_0} - q^2 \eta^{6v_0} - 2q^3 \eta^{6v_0} - q^4 \eta^{4v_0} - 4q^4 \eta^{6v_0} - 2q^5 \eta^{4v_0} - 4q^5 \eta^{6v_0} + q^5 \eta^{8v_0} \\
I(0, -2, -2, 1) &= \eta^{v_0} (-q^3 \eta^{5v_0} - q^3 \eta^{7v_0} - q^4 \eta^{5v_0} + 2q^5 \eta^{7v_0}) \\
I(0, -2, -2, 2) &= \eta^{2v_0} (q^5 \eta^{6v_0}) \\
I(0, 0, -2, -2) &= \eta^{-4v_0} (q^3 \eta^{4v_0} - 2q^4 \eta^{2v_0} - q^4 \eta^{4v_0} - 5q^5 \eta^{2v_0} - 4q^5 \eta^{4v_0}) \\
I(0, 0, -2, -1) &= \eta^{-3v_0} (q^2 \eta^{3v_0} + 2q^3 \eta^{3v_0} + q^3 \eta^{5v_0} - q^4 \eta^{v_0} - q^4 \eta^{3v_0} - 2q^5 \eta^{v_0} - 9q^5 \eta^{3v_0} - 4q^5 \eta^{5v_0}) \\
I(0, 0, -2, 0) &= \eta^{-2v_0} (q^2 \eta^{4v_0} + q^3 \eta^{2v_0} + 3q^3 \eta^{4v_0} + q^3 \eta^{6v_0} - q^4 \eta^{2v_0} + q^4 \eta^{4v_0} - q^4 \eta^{6v_0} - 4q^5 \eta^{2v_0} - 8q^5 \eta^{4v_0} - \\
I(0, 0, -2, 1) &= \eta^{-v_0} (q^2 \eta^{5v_0} + q^3 \eta^{3v_0} + 2q^3 \eta^{5v_0} - q^4 \eta^{5v_0} - q^4 \eta^{7v_0} - 4q^5 \eta^{3v_0} - 9q^5 \eta^{5v_0} - 2q^5 \eta^{7v_0}) \\
I(0, 0, -2, 2) &= q^3 \eta^{4v_0} - q^4 \eta^{4v_0} - 2q^4 \eta^{6v_0} - 4q^5 \eta^{4v_0} - 5q^5 \eta^{6v_0} \\
I(0, +2, -2, -2) &= \eta^{-6v_0} (q^5 \eta^{2v_0}) \\
I(0, +2, -2, -1) &= \eta^{-5v_0} (-q^3 \eta^{v_0} - q^3 \eta^{3v_0} - q^4 \eta^{3v_0} + 2q^5 \eta^{v_0}) \\
I(0, +2, -2, 0) &= \eta^{-4v_0} (q\eta^{2v_0} - q^2 \eta^{2v_0} - 2q^3 \eta^{2v_0} - 4q^4 \eta^{2v_0} - q^4 \eta^{4v_0} + q^5 - 4q^5 \eta^{2v_0} - 2q^5 \eta^{4v_0}) \\
I(0, +2, -2, 1) &= \eta^{-3v_0} (q^2 \eta^{3v_0} - q^3 \eta^{v_0} - 2q^4 \eta^{v_0} - 3q^4 \eta^{3v_0} - 2q^5 \eta^{v_0} - 8q^5 \eta^{3v_0} - q^5 \eta^{5v_0}) \\
I(0, +2, -2, 2) &= \eta^{-2v_0} (q^3 \eta^{4v_0} - 2q^4 \eta^{2v_0} - q^4 \eta^{4v_0} - 5q^5 \eta^{2v_0} - 4q^5 \eta^{4v_0}) \\
I(0, +4, -2, -1) &= \eta^{-7v_0} (-q^5 \eta^{v_0} - q^5 \eta^{3v_0}) \\
I(0, +4, -2, 0) &= \eta^{-6v_0} (q^2 \eta^{2v_0} - q^5 \eta^{2v_0}) \\
I(0, +4, -2, 1) &= \eta^{-5v_0} (q^4 \eta^{v_0} + q^4 \eta^{3v_0} + q^5 \eta^{v_0} + q^5 \eta^{3v_0}) \\
I(0, +4, -2, 2) &= \eta^{-4v_0} (q^5 \eta^{2v_0}) \\
I(0, -4, -1, -2) &= \eta^{v_0} (q^5 \eta^{3v_0}) \\
I(0, -4, -1, -1) &= \eta^{2v_0} (q^4 \eta^{2v_0} + q^4 \eta^{4v_0} + q^5 \eta^{2v_0} + q^5 \eta^{4v_0}) \\
I(0, -4, -1, 0) &= \eta^{3v_0} (q^2 \eta^{3v_0}) \\
I(0, -4, -1, 1) &= \eta^{4v_0} (-q^5 \eta^{2v_0} - q^5 \eta^{4v_0}) \\
I(0, -2, -1, -2) &= \eta^{-v_0} (q^2 \eta^{3v_0} + q^3 \eta^{v_0} + 2q^3 \eta^{3v_0} - q^4 \eta^{3v_0} - q^4 \eta^{5v_0} - 4q^5 \eta^{v_0} - 9q^5 \eta^{3v_0} - 2q^5 \eta^{5v_0}) \\
I(0, -2, -1, -1) &= q^2 \eta^{2v_0} + q^2 \eta^{4v_0} + q^3 \eta^{2v_0} - q^4 \eta^{2v_0} - 3q^4 \eta^{4v_0} - q^5 - 8q^5 \eta^{2v_0} - 8q^5 \eta^{4v_0} \\
I(0, -2, -1, 0) &= \eta^{v_0} (q\eta^{3v_0} - q^3 \eta^{3v_0} - q^4 \eta^{v_0} - 4q^4 \eta^{3v_0} - q^4 \eta^{5v_0} - 2q^5 \eta^{v_0} - 8q^5 \eta^{3v_0} - q^5 \eta^{5v_0}) \\
I(0, -2, -1, 1) &= \eta^{2v_0} (-q^3 \eta^{2v_0} - q^3 \eta^{4v_0} - q^4 \eta^{2v_0} - q^4 \eta^{4v_0} - q^5 \eta^{2v_0}) \\
I(0, -2, -1, 2) &= \eta^{3v_0} (q^5 \eta^{3v_0}) \\
I(0, 0, -1, -2) &= \eta^{-3v_0} (q^2 \eta^{v_0} + 2q^3 \eta^{v_0} + q^3 \eta^{3v_0} - q^4 \eta^{-v_0} - q^4 \eta^{v_0} - 2q^5 \eta^{-v_0} - 9q^5 \eta^{v_0} - 4q^5 \eta^{3v_0}) \\
I(0, 0, -1, -1) &= \eta^{-2v_0} (-q\eta^{2v_0} + q^2 + q^2 \eta^{2v_0} + 2q^3 + 6q^3 \eta^{2v_0} + q^3 \eta^{4v_0} - q^4 + 6q^4 \eta^{2v_0} - 9q^5 - 5q^5 \eta^{2v_0} - 4q^5 \eta^{v_0} - \\
I(0, 0, -1, 0) &= \eta^{-v_0} (-q\eta^{v_0} - q\eta^{3v_0} + q^2 \eta^{v_0} + q^2 \eta^{3v_0} + 6q^3 \eta^{v_0} + 6q^3 \eta^{3v_0} - q^4 \eta^{-v_0} + 6q^4 \eta^{v_0} + 6q^4 \eta^{3v_0} - q^4 \eta^{5v_0}
\end{aligned}$$

$$\begin{aligned}
I(0,0,-1,1) &= -q\eta^{2v_0} + q^2\eta^{2v_0} + q^2\eta^{4v_0} + q^3 + 6q^3\eta^{2v_0} + 2q^3\eta^{4v_0} + 6q^4\eta^{2v_0} - q^4\eta^{4v_0} - 4q^5 - 5q^5\eta^{2v_0} - 9q^5\eta^{4v_0} \\
I(0,0,-1,2) &= \eta^{v_0} (q^2\eta^{3v_0} + q^3\eta^{v_0} + 2q^3\eta^{3v_0} - q^4\eta^{3v_0} - q^4\eta^{5v_0} - 4q^5\eta^{v_0} - 9q^5\eta^{3v_0} - 2q^5\eta^{5v_0}) \\
I(0,+2,-1,-2) &= \eta^{-5v_0} (q^5\eta^{v_0}) \\
I(0,+2,-1,-1) &= \eta^{-4v_0} (-q^3 - q^3\eta^{2v_0} - q^4 - q^4\eta^{2v_0} - q^5\eta^{2v_0}) \\
I(0,+2,-1,0) &= \eta^{-3v_0} (q\eta^{v_0} - q^3\eta^{v_0} - q^4\eta^{-v_0} - 4q^4\eta^{v_0} - q^4\eta^{3v_0} - q^5\eta^{-v_0} - 8q^5\eta^{v_0} - 2q^5\eta^{3v_0}) \\
I(0,+2,-1,1) &= \eta^{-2v_0} (q^2 + q^2\eta^{2v_0} + q^3\eta^{2v_0} - 3q^4 - q^4\eta^{2v_0} - 8q^5 - 8q^5\eta^{2v_0} - q^5\eta^{4v_0}) \\
I(0,+2,-1,2) &= \eta^{-v_0} (q^2\eta^{v_0} + 2q^3\eta^{v_0} + q^3\eta^{3v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - 2q^5\eta^{-v_0} - 9q^5\eta^{v_0} - 4q^5\eta^{3v_0}) \\
I(0,+4,-1,-1) &= \eta^{-6v_0} (-q^5 - q^5\eta^{2v_0}) \\
I(0,+4,-1,0) &= \eta^{-5v_0} (q^2\eta^{v_0}) \\
I(0,+4,-1,1) &= \eta^{-4v_0} (q^4 + q^4\eta^{2v_0} + q^5 + q^5\eta^{2v_0}) \\
I(0,+4,-1,2) &= \eta^{-3v_0} (q^5\eta^{v_0}) \\
I(0,-4,0,-2) &= \eta^{2v_0} (q^5) \\
I(0,-4,0,-1) &= \eta^{3v_0} (q^4\eta^{-v_0} + q^4\eta^{v_0} + q^5\eta^{-v_0} + q^5\eta^{v_0}) \\
I(0,-4,0,0) &= \eta^{4v_0} (q^2) \\
I(0,-4,0,1) &= \eta^{5v_0} (-q^5\eta^{-v_0} - q^5\eta^{v_0}) \\
I(0,-2,0,-2) &= q^2 + q^3\eta^{-2v_0} + 3q^3 + q^3\eta^{2v_0} - q^4\eta^{-2v_0} + q^4 - q^4\eta^{2v_0} - 4q^5\eta^{-2v_0} - 8q^5 - 4q^5\eta^{2v_0} \\
I(0,-2,0,-1) &= \eta^{v_0} (q^2\eta^{-v_0} + q^2\eta^{v_0} + q^3\eta^{-v_0} + q^3\eta^{v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - q^5\eta^{-3v_0} - 8q^5\eta^{-v_0} - 8q^5\eta^{v_0} - q^5\eta^{3v_0}) \\
I(0,-2,0,0) &= \eta^{2v_0} (q - q^4\eta^{-2v_0} - 4q^4 - q^4\eta^{2v_0} - 2q^5\eta^{-2v_0} - 8q^5 - 2q^5\eta^{2v_0}) \\
I(0,-2,0,1) &= \eta^{3v_0} (-q^3\eta^{-v_0} - q^3\eta^{v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - q^5\eta^{-v_0} - q^5\eta^{v_0}) \\
I(0,-2,0,2) &= \eta^{4v_0} (q^5) \\
I(0,0,0,-2) &= \eta^{-2v_0} (q^2 + q^3\eta^{-2v_0} + 3q^3 + q^3\eta^{2v_0} - q^4\eta^{-2v_0} + q^4 - q^4\eta^{2v_0} - 4q^5\eta^{-2v_0} - 8q^5 - 4q^5\eta^{2v_0}) \\
I(0,0,0,-1) &= \eta^{-v_0} (-q\eta^{-v_0} - q\eta^{v_0} + q^2\eta^{-v_0} + q^2\eta^{v_0} + 6q^3\eta^{-v_0} + 6q^3\eta^{v_0} - q^4\eta^{-3v_0} + 6q^4\eta^{-v_0} + 6q^4\eta^{v_0} - q^4\eta^{3v_0}) \\
I(0,0,0,0) &= 1 - 3q + q^2\eta^{-2v_0} - q^2 + q^2\eta^{2v_0} + 3q^3\eta^{-2v_0} + 10q^3 + 3q^3\eta^{2v_0} + q^4\eta^{-2v_0} + 14q^4 + q^4\eta^{2v_0} - 8q^5 \\
I(0,0,0,1) &= \eta^{v_0} (-q\eta^{-v_0} - q\eta^{v_0} + q^2\eta^{-v_0} + q^2\eta^{v_0} + 6q^3\eta^{-v_0} + 6q^3\eta^{v_0} - q^4\eta^{-3v_0} + 6q^4\eta^{-v_0} + 6q^4\eta^{v_0} - q^4\eta^{3v_0}) \\
I(0,0,0,2) &= \eta^{2v_0} (q^2 + q^3\eta^{-2v_0} + 3q^3 + q^3\eta^{2v_0} - q^4\eta^{-2v_0} + q^4 - q^4\eta^{2v_0} - 4q^5\eta^{-2v_0} - 8q^5 - 4q^5\eta^{2v_0}) \\
I(0,+2,0,-2) &= \eta^{-4v_0} (q^5) \\
I(0,+2,0,-1) &= \eta^{-3v_0} (-q^3\eta^{-v_0} - q^3\eta^{v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - q^5\eta^{-v_0} - q^5\eta^{v_0}) \\
I(0,+2,0,0) &= \eta^{-2v_0} (q - q^4\eta^{-2v_0} - 4q^4 - q^4\eta^{2v_0} - 2q^5\eta^{-2v_0} - 8q^5 - 2q^5\eta^{2v_0}) \\
I(0,+2,0,1) &= \eta^{-v_0} (q^2\eta^{-v_0} + q^2\eta^{v_0} + q^3\eta^{-v_0} + q^3\eta^{v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - q^5\eta^{-3v_0} - 8q^5\eta^{-v_0} - 8q^5\eta^{v_0} - q^5\eta^{3v_0}) \\
I(0,+2,0,2) &= q^2 + q^3\eta^{-2v_0} + 3q^3 + q^3\eta^{2v_0} - q^4\eta^{-2v_0} + q^4 - q^4\eta^{2v_0} - 4q^5\eta^{-2v_0} - 8q^5 - 4q^5\eta^{2v_0} \\
I(0,+4,0,-1) &= \eta^{-5v_0} (-q^5\eta^{-v_0} - q^5\eta^{v_0}) \\
I(0,+4,0,0) &= \eta^{-4v_0} (q^2) \\
I(0,+4,0,1) &= \eta^{-3v_0} (q^4\eta^{-v_0} + q^4\eta^{v_0} + q^5\eta^{-v_0} + q^5\eta^{v_0}) \\
I(0,+4,0,2) &= \eta^{-2v_0} (q^5) \\
I(0,-4,1,-2) &= \eta^{3v_0} (q^5\eta^{-3v_0}) \\
I(0,-4,1,-1) &= \eta^{4v_0} (q^4\eta^{-4v_0} + q^4\eta^{-2v_0} + q^5\eta^{-4v_0} + q^5\eta^{-2v_0}) \\
I(0,-4,1,0) &= \eta^{5v_0} (q^2\eta^{-3v_0}) \\
I(0,-4,1,1) &= \eta^{6v_0} (-q^5\eta^{-4v_0} - q^5\eta^{-2v_0}) \\
I(0,-2,1,-2) &= \eta^{v_0} (q^2\eta^{-3v_0} + 2q^3\eta^{-3v_0} + q^3\eta^{-v_0} - q^4\eta^{-5v_0} - q^4\eta^{-3v_0} - 2q^5\eta^{-5v_0} - 9q^5\eta^{-3v_0} - 4q^5\eta^{-v_0}) \\
I(0,-2,1,-1) &= \eta^{2v_0} (q^2\eta^{-4v_0} + q^2\eta^{-2v_0} + q^3\eta^{-2v_0} - 3q^4\eta^{-4v_0} - q^4\eta^{-2v_0} - 8q^5\eta^{-4v_0} - 8q^5\eta^{-2v_0} - q^5) \\
I(0,-2,1,0) &= \eta^{3v_0} (q\eta^{-3v_0} - q^3\eta^{-3v_0} - q^4\eta^{-5v_0} - 4q^4\eta^{-3v_0} - q^4\eta^{-v_0} - q^5\eta^{-5v_0} - 8q^5\eta^{-3v_0} - 2q^5\eta^{-v_0})
\end{aligned}$$

$$\begin{aligned}
I(0, -2, 1, 1) &= \eta^{4v_0} (-q^3\eta^{-4v_0} - q^3\eta^{-2v_0} - q^4\eta^{-4v_0} - q^4\eta^{-2v_0} - q^5\eta^{-2v_0}) \\
I(0, -2, 1, 2) &= \eta^{5v_0} (q^5\eta^{-3v_0}) \\
I(0, 0, 1, -2) &= \eta^{-v_0} (q^2\eta^{-v_0} + q^3\eta^{-3v_0} + 2q^3\eta^{-v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - 4q^5\eta^{-3v_0} - 9q^5\eta^{-v_0} - 2q^5\eta^{v_0}) \\
I(0, 0, 1, -1) &= -q\eta^{-2v_0} + q^2\eta^{-2v_0} + q^2 + q^3\eta^{-4v_0} + 6q^3\eta^{-2v_0} + 2q^3 + 6q^4\eta^{-2v_0} - q^4 - 4q^5\eta^{-4v_0} - 5q^5\eta^{-2v_0} \\
I(0, 0, 1, 0) &= \eta^{v_0} (-q\eta^{-3v_0} - q\eta^{-v_0} + q^2\eta^{-3v_0} + q^2\eta^{-v_0} + 6q^3\eta^{-3v_0} + 6q^3\eta^{-v_0} - q^4\eta^{-5v_0} + 6q^4\eta^{-3v_0} + 6q^4\eta^{-v_0}) \\
I(0, 0, 1, 1) &= \eta^{2v_0} (-q\eta^{-2v_0} + q^2\eta^{-4v_0} + q^2\eta^{-2v_0} + 2q^3\eta^{-4v_0} + 6q^3\eta^{-2v_0} + q^3 - q^4\eta^{-4v_0} + 6q^4\eta^{-2v_0} - 9q^5\eta^{-v_0}) \\
I(0, 0, 1, 2) &= \eta^{3v_0} (q^2\eta^{-3v_0} + 2q^3\eta^{-3v_0} + q^3\eta^{-v_0} - q^4\eta^{-5v_0} - q^4\eta^{-3v_0} - 2q^5\eta^{-5v_0} - 9q^5\eta^{-3v_0} - 4q^5\eta^{-v_0}) \\
I(0, +2, 1, -2) &= \eta^{-3v_0} (q^5\eta^{-v_0}) \\
I(0, +2, 1, -1) &= \eta^{-2v_0} (-q^3\eta^{-2v_0} - q^3 - q^4\eta^{-2v_0} - q^4 - q^5\eta^{-2v_0}) \\
I(0, +2, 1, 0) &= \eta^{-v_0} (q\eta^{-v_0} - q^3\eta^{-v_0} - q^4\eta^{-3v_0} - 4q^4\eta^{-v_0} - q^4\eta^{v_0} - 2q^5\eta^{-3v_0} - 8q^5\eta^{-v_0} - q^5\eta^{v_0}) \\
I(0, +2, 1, 1) &= q^2\eta^{-2v_0} + q^2 + q^3\eta^{-2v_0} - q^4\eta^{-2v_0} - 3q^4 - q^5\eta^{-4v_0} - 8q^5\eta^{-2v_0} - 8q^5 \\
I(0, +2, 1, 2) &= \eta^{v_0} (q^2\eta^{-v_0} + q^3\eta^{-3v_0} + 2q^3\eta^{-v_0} - q^4\eta^{-v_0} - q^4\eta^{v_0} - 4q^5\eta^{-3v_0} - 9q^5\eta^{-v_0} - 2q^5\eta^{v_0}) \\
I(0, +4, 1, -1) &= \eta^{-4v_0} (-q^5\eta^{-2v_0} - q^5) \\
I(0, +4, 1, 0) &= \eta^{-3v_0} (q^2\eta^{-v_0}) \\
I(0, +4, 1, 1) &= \eta^{-2v_0} (q^4\eta^{-2v_0} + q^4 + q^5\eta^{-2v_0} + q^5) \\
I(0, +4, 1, 2) &= \eta^{-v_0} (q^5\eta^{-v_0}) \\
I(0, -4, 2, -2) &= \eta^{4v_0} (q^5\eta^{-6v_0}) \\
I(0, -4, 2, -1) &= \eta^{5v_0} (q^4\eta^{-7v_0} + q^4\eta^{-5v_0} + q^5\eta^{-7v_0} + q^5\eta^{-5v_0}) \\
I(0, -4, 2, 0) &= \eta^{6v_0} (q^2\eta^{-6v_0} - q^5\eta^{-6v_0}) \\
I(0, -4, 2, 1) &= \eta^{7v_0} (-q^5\eta^{-7v_0} - q^5\eta^{-5v_0}) \\
I(0, -2, 2, -2) &= \eta^{2v_0} (q^3\eta^{-4v_0} - 2q^4\eta^{-6v_0} - q^4\eta^{-4v_0} - 5q^5\eta^{-6v_0} - 4q^5\eta^{-4v_0}) \\
I(0, -2, 2, -1) &= \eta^{3v_0} (q^2\eta^{-5v_0} - q^3\eta^{-7v_0} - 2q^4\eta^{-7v_0} - 3q^4\eta^{-5v_0} - 2q^5\eta^{-7v_0} - 8q^5\eta^{-5v_0} - q^5\eta^{-3v_0}) \\
I(0, -2, 2, 0) &= \eta^{4v_0} (q\eta^{-6v_0} - q^2\eta^{-6v_0} - 2q^3\eta^{-6v_0} - 4q^4\eta^{-6v_0} - q^4\eta^{-4v_0} + q^5\eta^{-8v_0} - 4q^5\eta^{-6v_0} - 2q^5\eta^{-4v_0}) \\
I(0, -2, 2, 1) &= \eta^{5v_0} (-q^3\eta^{-7v_0} - q^3\eta^{-5v_0} - q^4\eta^{-5v_0} + 2q^5\eta^{-7v_0}) \\
I(0, -2, 2, 2) &= \eta^{6v_0} (q^5\eta^{-6v_0}) \\
I(0, 0, 2, -2) &= q^3\eta^{-4v_0} - q^4\eta^{-4v_0} - 2q^4\eta^{-2v_0} - 4q^5\eta^{-4v_0} - 5q^5\eta^{-2v_0} \\
I(0, 0, 2, -1) &= \eta^{v_0} (q^2\eta^{-3v_0} + q^3\eta^{-5v_0} + 2q^3\eta^{-3v_0} - q^4\eta^{-3v_0} - q^4\eta^{-v_0} - 4q^5\eta^{-5v_0} - 9q^5\eta^{-3v_0} - 2q^5\eta^{-v_0}) \\
I(0, 0, 2, 0) &= \eta^{2v_0} (q^2\eta^{-4v_0} + q^3\eta^{-6v_0} + 3q^3\eta^{-4v_0} + q^3\eta^{-2v_0} - q^4\eta^{-6v_0} + q^4\eta^{-4v_0} - q^4\eta^{-2v_0} - 4q^5\eta^{-6v_0} - 9q^5\eta^{-4v_0} - 2q^5\eta^{-2v_0}) \\
I(0, 0, 2, 1) &= \eta^{3v_0} (q^2\eta^{-5v_0} + 2q^3\eta^{-5v_0} + q^3\eta^{-3v_0} - q^4\eta^{-7v_0} - q^4\eta^{-5v_0} - 2q^5\eta^{-7v_0} - 9q^5\eta^{-5v_0} - 4q^5\eta^{-3v_0}) \\
I(0, 0, 2, 2) &= \eta^{4v_0} (q^3\eta^{-4v_0} - 2q^4\eta^{-6v_0} - q^4\eta^{-4v_0} - 5q^5\eta^{-6v_0} - 4q^5\eta^{-4v_0}) \\
I(0, +2, 2, -2) &= \eta^{-2v_0} (q^5\eta^{-2v_0}) \\
I(0, +2, 2, -1) &= \eta^{-v_0} (-q^3\eta^{-3v_0} - q^3\eta^{-v_0} - q^4\eta^{-3v_0} + 2q^5\eta^{-v_0}) \\
I(0, +2, 2, 0) &= q\eta^{-2v_0} - q^2\eta^{-2v_0} - 2q^3\eta^{-2v_0} - q^4\eta^{-4v_0} - 4q^4\eta^{-2v_0} - 2q^5\eta^{-4v_0} - 4q^5\eta^{-2v_0} + q^5 \\
I(0, +2, 2, 1) &= \eta^{v_0} (q^2\eta^{-3v_0} - q^3\eta^{-v_0} - 3q^4\eta^{-3v_0} - 2q^4\eta^{-v_0} - q^5\eta^{-5v_0} - 8q^5\eta^{-3v_0} - 2q^5\eta^{-v_0}) \\
I(0, +2, 2, 2) &= \eta^{2v_0} (q^3\eta^{-4v_0} - q^4\eta^{-4v_0} - 2q^4\eta^{-2v_0} - 4q^5\eta^{-4v_0} - 5q^5\eta^{-2v_0}) \\
I(0, +4, 2, -1) &= \eta^{-3v_0} (-q^5\eta^{-3v_0} - q^5\eta^{-v_0}) \\
I(0, +4, 2, 0) &= \eta^{-2v_0} (q^2\eta^{-2v_0} - q^5\eta^{-2v_0}) \\
I(0, +4, 2, 1) &= \eta^{-v_0} (q^4\eta^{-3v_0} + q^4\eta^{-v_0} + q^5\eta^{-3v_0} + q^5\eta^{-v_0}) \\
I(0, +4, 2, 2) &= q^5\eta^{-2v_0} \\
I(+2, -4, -2, 2) &= \eta^{2v_0} (q^5\eta^{12v_0}) \\
I(+2, -2, -2, -1) &= \eta^{-3v_0} (q^5\eta^{9v_0})
\end{aligned}$$

$$\begin{aligned}
I(+2, -2, -2, 0) &= \eta^{-2v_0} (q^4 \eta^{10v_0} + q^5 \eta^{10v_0}) \\
I(+2, -2, -2, 1) &= \eta^{-v_0} (q^3 \eta^{11v_0} + q^4 \eta^{11v_0} + q^5 \eta^{11v_0}) \\
I(+2, -2, -2, 2) &= q^2 \eta^{12v_0} - 2q^5 \eta^{12v_0} \\
I(+2, 0, -2, -2) &= \eta^{-6v_0} (q^5 \eta^{6v_0}) \\
I(+2, 0, -2, -1) &= \eta^{-5v_0} (q^5 \eta^{7v_0}) \\
I(+2, 0, -2, 0) &= \eta^{-4v_0} (q^5 \eta^{8v_0}) \\
I(+2, 0, -2, 1) &= \eta^{-3v_0} (q^5 \eta^{9v_0}) \\
I(+2, 0, -2, 2) &= \eta^{-2v_0} (q^5 \eta^{10v_0}) \\
I(+2, +2, -2, -2) &= \eta^{-8v_0} (q^2 \eta^{4v_0} - 2q^5 \eta^{4v_0}) \\
I(+2, +2, -2, -1) &= \eta^{-7v_0} (q^3 \eta^{5v_0} + q^4 \eta^{5v_0} + q^5 \eta^{5v_0}) \\
I(+2, +2, -2, 0) &= \eta^{-6v_0} (q^4 \eta^{6v_0} + q^5 \eta^{6v_0}) \\
I(+2, +2, -2, 1) &= \eta^{-5v_0} (q^5 \eta^{7v_0}) \\
I(+2, +4, -2, -2) &= \eta^{-10v_0} (q^5 \eta^{4v_0}) \\
I(+2, -2, -1, 0) &= \eta^{-v_0} (q^5 \eta^{7v_0} + q^5 \eta^{9v_0}) \\
I(+2, -2, -1, 1) &= q^4 \eta^{8v_0} + q^4 \eta^{10v_0} + 2q^5 \eta^{8v_0} + q^5 \eta^{10v_0} \\
I(+2, -2, -1, 2) &= \eta^{v_0} (q^3 \eta^{9v_0} + q^4 \eta^{9v_0} + q^5 \eta^{9v_0}) \\
I(+2, 0, -1, -2) &= \eta^{-5v_0} (-q^3 \eta^{3v_0} - q^3 \eta^{5v_0} - q^4 \eta^{5v_0} + 2q^5 \eta^{3v_0}) \\
I(+2, 0, -1, -1) &= \eta^{-4v_0} (-q^3 \eta^{4v_0} - q^3 \eta^{6v_0} - q^4 \eta^{4v_0} - q^4 \eta^{6v_0} - q^5 \eta^{6v_0}) \\
I(+2, 0, -1, 0) &= \eta^{-3v_0} (-q^3 \eta^{5v_0} - q^3 \eta^{7v_0} - q^4 \eta^{5v_0} - q^4 \eta^{7v_0} - q^5 \eta^{5v_0} - q^5 \eta^{7v_0}) \\
I(+2, 0, -1, 1) &= \eta^{-2v_0} (-q^3 \eta^{6v_0} - q^3 \eta^{8v_0} - q^4 \eta^{6v_0} - q^4 \eta^{8v_0} - q^5 \eta^{6v_0}) \\
I(+2, 0, -1, 2) &= \eta^{-v_0} (-q^3 \eta^{7v_0} - q^3 \eta^{9v_0} - q^4 \eta^{7v_0} + 2q^5 \eta^{9v_0}) \\
I(+2, +2, -1, -2) &= \eta^{-7v_0} (q^3 \eta^{3v_0} + q^4 \eta^{3v_0} + q^5 \eta^{3v_0}) \\
I(+2, +2, -1, -1) &= \eta^{-6v_0} (q^4 \eta^{2v_0} + q^4 \eta^{4v_0} + q^5 \eta^{2v_0} + 2q^5 \eta^{4v_0}) \\
I(+2, +2, -1, 0) &= \eta^{-5v_0} (q^5 \eta^{3v_0} + q^5 \eta^{5v_0}) \\
I(+2, -2, 0, -2) &= \eta^{-2v_0} (-q^4 \eta^{4v_0} - 2q^5 \eta^{4v_0}) \\
I(+2, -2, 0, -1) &= \eta^{-v_0} (-q^4 \eta^{5v_0} - 2q^5 \eta^{5v_0}) \\
I(+2, -2, 0, 0) &= -q^4 \eta^{6v_0} - q^5 \eta^{6v_0} \\
I(+2, -2, 0, 1) &= \eta^{v_0} (q^5 \eta^{5v_0} + q^5 \eta^{7v_0}) \\
I(+2, -2, 0, 2) &= \eta^{2v_0} (q^4 \eta^{6v_0} + q^5 \eta^{6v_0}) \\
I(+2, 0, 0, -2) &= \eta^{-4v_0} (q \eta^{2v_0} - q^2 \eta^{2v_0} - 2q^3 \eta^{2v_0} - 4q^4 \eta^{2v_0} - q^4 \eta^{4v_0} + q^5 - 4q^5 \eta^{2v_0} - 2q^5 \eta^{4v_0}) \\
I(+2, 0, 0, -1) &= \eta^{-3v_0} (q \eta^{3v_0} - q^3 \eta^{3v_0} - q^4 \eta^{v_0} - 4q^4 \eta^{3v_0} - q^4 \eta^{5v_0} - q^5 \eta^{v_0} - 8q^5 \eta^{3v_0} - 2q^5 \eta^{5v_0}) \\
I(+2, 0, 0, 0) &= \eta^{-2v_0} (q \eta^{4v_0} - q^4 \eta^{2v_0} - 4q^4 \eta^{4v_0} - q^4 \eta^{6v_0} - 2q^5 \eta^{2v_0} - 8q^5 \eta^{4v_0} - 2q^5 \eta^{6v_0}) \\
I(+2, 0, 0, 1) &= \eta^{-v_0} (q \eta^{5v_0} - q^3 \eta^{5v_0} - q^4 \eta^{3v_0} - 4q^4 \eta^{5v_0} - q^4 \eta^{7v_0} - 2q^5 \eta^{3v_0} - 8q^5 \eta^{5v_0} - q^5 \eta^{7v_0}) \\
I(+2, 0, 0, 2) &= q \eta^{6v_0} - q^2 \eta^{6v_0} - 2q^3 \eta^{6v_0} - q^4 \eta^{4v_0} - 4q^4 \eta^{6v_0} - 2q^5 \eta^{4v_0} - 4q^5 \eta^{6v_0} + q^5 \eta^{8v_0} \\
I(+2, +2, 0, -2) &= \eta^{-6v_0} (q^4 \eta^{2v_0} + q^5 \eta^{2v_0}) \\
I(+2, +2, 0, -1) &= \eta^{-5v_0} (q^5 \eta^{v_0} + q^5 \eta^{3v_0}) \\
I(+2, +2, 0, 0) &= \eta^{-4v_0} (-q^4 \eta^{2v_0} - q^5 \eta^{2v_0}) \\
I(+2, +2, 0, 1) &= \eta^{-3v_0} (-q^4 \eta^{3v_0} - 2q^5 \eta^{3v_0}) \\
I(+2, +2, 0, 2) &= \eta^{-2v_0} (-q^4 \eta^{4v_0} - 2q^5 \eta^{4v_0}) \\
I(+2, -2, 1, -2) &= \eta^{-v_0} (-q^4 \eta^{v_0} - 4q^5 \eta^{v_0} - 2q^5 \eta^{3v_0}) \\
I(+2, -2, 1, -1) &= -q^4 \eta^{2v_0} - 3q^5 \eta^{2v_0} - q^5 \eta^{4v_0}
\end{aligned}$$

$$\begin{aligned}
I(+2, -2, 1, 0) &= \eta^{v_0} (-q^4 \eta^{3v_0} - 2q^5 \eta^{3v_0}) \\
I(+2, -2, 1, 2) &= \eta^{3v_0} (q^5 \eta^{3v_0}) \\
I(+2, 0, 1, -2) &= \eta^{-3v_0} (q^2 \eta^{v_0} - q^3 \eta^{-v_0} - 2q^4 \eta^{-v_0} - 3q^4 \eta^{v_0} - 2q^5 \eta^{-v_0} - 8q^5 \eta^{v_0} - q^5 \eta^{3v_0}) \\
I(+2, 0, 1, -1) &= \eta^{-2v_0} (q^2 + q^2 \eta^{2v_0} + q^3 \eta^{2v_0} - 3q^4 - q^4 \eta^{2v_0} - 8q^5 - 8q^5 \eta^{2v_0} - q^5 \eta^{4v_0}) \\
I(+2, 0, 1, 0) &= \eta^{-v_0} (q^2 \eta^{v_0} + q^2 \eta^{3v_0} + q^3 \eta^{v_0} + q^3 \eta^{3v_0} - q^4 \eta^{v_0} - q^4 \eta^{3v_0} - q^5 \eta^{-v_0} - 8q^5 \eta^{v_0} - 8q^5 \eta^{3v_0} - q^5 \eta^{5v_0}) \\
I(+2, 0, 1, 1) &= q^2 \eta^{2v_0} + q^2 \eta^{4v_0} + q^3 \eta^{2v_0} - q^4 \eta^{2v_0} - 3q^4 \eta^{4v_0} - q^5 - 8q^5 \eta^{2v_0} - 8q^5 \eta^{4v_0} \\
I(+2, 0, 1, 2) &= \eta^{v_0} (q^2 \eta^{3v_0} - q^3 \eta^{5v_0} - 3q^4 \eta^{3v_0} - 2q^4 \eta^{5v_0} - q^5 \eta^{v_0} - 8q^5 \eta^{3v_0} - 2q^5 \eta^{5v_0}) \\
I(+2, +2, 1, -2) &= \eta^{-5v_0} (q^5 \eta^{v_0}) \\
I(+2, +2, 1, 0) &= \eta^{-3v_0} (-q^4 \eta^{v_0} - 2q^5 \eta^{v_0}) \\
I(+2, +2, 1, 1) &= \eta^{-2v_0} (-q^4 \eta^{2v_0} - q^5 - 3q^5 \eta^{2v_0}) \\
I(+2, +2, 1, 2) &= \eta^{-v_0} (-q^4 \eta^{3v_0} - 2q^5 \eta^{v_0} - 4q^5 \eta^{3v_0}) \\
I(+2, -2, 2, -2) &= q^3 - 2q^4 \eta^{-2v_0} - q^4 - 5q^5 \eta^{-2v_0} - 4q^5 \\
I(+2, -2, 2, -1) &= \eta^{v_0} (-q^4 \eta^{-v_0} - 4q^5 \eta^{-v_0} - 2q^5 \eta^{v_0}) \\
I(+2, -2, 2, 0) &= \eta^{2v_0} (-q^4 - 2q^5) \\
I(+2, 0, 2, -2) &= \eta^{-2v_0} (q^3 - 2q^4 \eta^{-2v_0} - q^4 - 5q^5 \eta^{-2v_0} - 4q^5) \\
I(+2, 0, 2, -1) &= \eta^{-v_0} (q^2 \eta^{-v_0} + 2q^3 \eta^{-v_0} + q^3 \eta^{v_0} - q^4 \eta^{-3v_0} - q^4 \eta^{-v_0} - 2q^5 \eta^{-3v_0} - 9q^5 \eta^{-v_0} - 4q^5 \eta^{v_0}) \\
I(+2, 0, 2, 0) &= q^2 + q^3 \eta^{-2v_0} + 3q^3 + q^3 \eta^{2v_0} - q^4 \eta^{-2v_0} + q^4 - q^4 \eta^{2v_0} - 4q^5 \eta^{-2v_0} - 8q^5 - 4q^5 \eta^{2v_0} \\
I(+2, 0, 2, 1) &= \eta^{v_0} (q^2 \eta^{v_0} + q^3 \eta^{-v_0} + 2q^3 \eta^{v_0} - q^4 \eta^{v_0} - q^4 \eta^{3v_0} - 4q^5 \eta^{-v_0} - 9q^5 \eta^{v_0} - 2q^5 \eta^{3v_0}) \\
I(+2, 0, 2, 2) &= \eta^{2v_0} (q^3 - q^4 - 2q^4 \eta^{2v_0} - 4q^5 - 5q^5 \eta^{2v_0}) \\
I(+2, +2, 2, 0) &= \eta^{-2v_0} (-q^4 - 2q^5) \\
I(+2, +2, 2, 1) &= \eta^{-v_0} (-q^4 \eta^{v_0} - 2q^5 \eta^{-v_0} - 4q^5 \eta^{v_0}) \\
I(+2, +2, 2, 2) &= q^3 - q^4 - 2q^4 \eta^{2v_0} - 4q^5 - 5q^5 \eta^{2v_0} \\
I(+4, -2, -2, 2) &= \eta^{-2v_0} (q^5 \eta^{16v_0}) \\
I(+4, +2, -2, -2) &= \eta^{-10v_0} (q^5 \eta^{8v_0}) \\
I(+4, 0, -1, -2) &= \eta^{-7v_0} (-q^5 \eta^{7v_0} - q^5 \eta^{9v_0}) \\
I(+4, 0, -1, -1) &= \eta^{-6v_0} (-q^5 \eta^{8v_0} - q^5 \eta^{10v_0}) \\
I(+4, 0, -1, 0) &= \eta^{-5v_0} (-q^5 \eta^{9v_0} - q^5 \eta^{11v_0}) \\
I(+4, 0, -1, 1) &= \eta^{-4v_0} (-q^5 \eta^{10v_0} - q^5 \eta^{12v_0}) \\
I(+4, 0, -1, 2) &= \eta^{-3v_0} (-q^5 \eta^{11v_0} - q^5 \eta^{13v_0}) \\
I(+4, 0, 0, -2) &= \eta^{-6v_0} (q^2 \eta^{6v_0} - q^5 \eta^{6v_0}) \\
I(+4, 0, 0, -1) &= \eta^{-5v_0} (q^2 \eta^{7v_0}) \\
I(+4, 0, 0, 0) &= \eta^{-4v_0} (q^2 \eta^{8v_0}) \\
I(+4, 0, 0, 1) &= \eta^{-3v_0} (q^2 \eta^{9v_0}) \\
I(+4, 0, 0, 2) &= \eta^{-2v_0} (q^2 \eta^{10v_0} - q^5 \eta^{10v_0}) \\
I(+4, 0, 1, -2) &= \eta^{-5v_0} (q^4 \eta^{3v_0} + q^4 \eta^{5v_0} + q^5 \eta^{3v_0} + q^5 \eta^{5v_0}) \\
I(+4, 0, 1, -1) &= \eta^{-4v_0} (q^4 \eta^{4v_0} + q^4 \eta^{6v_0} + q^5 \eta^{4v_0} + q^5 \eta^{6v_0}) \\
I(+4, 0, 1, 0) &= \eta^{-3v_0} (q^4 \eta^{5v_0} + q^4 \eta^{7v_0} + q^5 \eta^{5v_0} + q^5 \eta^{7v_0}) \\
I(+4, 0, 1, 1) &= \eta^{-2v_0} (q^4 \eta^{6v_0} + q^4 \eta^{8v_0} + q^5 \eta^{6v_0} + q^5 \eta^{8v_0}) \\
I(+4, 0, 1, 2) &= \eta^{-v_0} (q^4 \eta^{7v_0} + q^4 \eta^{9v_0} + q^5 \eta^{7v_0} + q^5 \eta^{9v_0}) \\
I(+4, 0, 2, -2) &= \eta^{-4v_0} (q^5 \eta^{2v_0}) \\
I(+4, 0, 2, -1) &= \eta^{-3v_0} (q^5 \eta^{3v_0})
\end{aligned}$$

$$I(+4, 0, 2, 0) = \eta^{-2v_0} (q^5 \eta^{4v_0})$$

$$I(+4, 0, 2, 1) = \eta^{-v_0} (q^5 \eta^{5v_0})$$

$$I(+4, 0, 2, 2) = q^5 \eta^{6v_0}$$